

# Logbook — Anomalous transport in soaring flights

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Working log

*Working notebook. The timeline is regenerated from the git history; the narrative notes are written by hand.*

## 1 Timeline (auto-generated from git)

| Date       | Event (commit subject)                                                              |
|------------|-------------------------------------------------------------------------------------|
| 2026-08-04 | docs(plan): add the Chapter 3 inventory, which overrules the earlier sketches       |
| 2026-08-04 | fix(audit): test the arrangement of the residuals, not only their size              |
| 2026-08-04 | feat(dataset): read the discards as a cascade, and check them against the calendar  |
| 2026-08-04 | refactor(analysis): separate computing from drawing, split the grab-bag module      |
| 2026-08-04 | chore: drop what the repo no longer needs; break the oversized parameter table      |
| 2026-08-04 | chore: regenerate the committed artefacts                                           |
| 2026-08-04 | docs(logbook): record the pipeline build and the MSD audit, realign Next steps      |
| 2026-08-04 | thesis(ch3): report the two MSD estimators as two measurements, not one             |
| 2026-08-04 | thesis(ch2): the pipeline as built, the analysis ensemble, and Sec. 2.8 in full     |
| 2026-08-04 | docs: the on-disk layout, the regeneration order and the handoff to the MSD         |
| 2026-08-04 | feat(prelim): the preliminary characterization of Sec. 2.8, on real maps            |
| 2026-08-04 | feat(audit): keep what the MSD averages away, and use it to test the curve          |
| 2026-08-04 | feat(reporting): the pipeline census, the macro contract and the regeneration chain |
| 2026-08-04 | feat(verify): check the written tables against the invariants, and against the code |
| 2026-08-04 | feat(transport): the two mean-squared-displacement estimators                       |
| 2026-08-04 | feat(analysis): stream whole flights out of the fix table                           |
| 2026-08-04 | feat(preproc): the seven-stage trajectory pipeline                                  |
| 2026-08-04 | feat(igc): tolerate partial altitude fields and the midnight roll-over              |
| 2026-08-04 | chore(tooling): require Python 3.12, fix the mypy roots, sweep the formatting       |
| 2026-08-03 | chore: regenerate the committed artefacts                                           |
| 2026-08-03 | docs: align the pipeline guide, complete the bibliography inventory                 |
| 2026-08-03 | docs(thesis): apply the second review pass                                          |
| 2026-08-03 | feat(census): count the placeholder dates, per season                               |
| 2026-08-03 | feat(preproc): altitude-activity cut 75 -> 600 m                                    |
| 2026-08-03 | fix(analysis): mark the gap cut at the bound actually applied                       |
| 2026-08-03 | feat(analysis): draw the third flight-level diagnostic                              |

| Date       | Event (commit subject)                                                                                    |
|------------|-----------------------------------------------------------------------------------------------------------|
| 2026-08-03 | fix(analysis): judge barometric presence strictly, from one definition                                    |
| 2026-07-24 | docs(thesis): review-pass revisions, geodesy appendix and pipeline map                                    |
| 2026-07-24 | docs(preproc): align pipeline guide with 20 km, \Preproc macros and the contract                          |
| 2026-07-24 | feat(preproc): 20 km path cut; export thresholds and census as macros                                     |
| 2026-07-13 | docs(thesis): trim next-steps plan; sync logbook                                                          |
| 2026-07-13 | docs(thesis): running headers and typographic refinements                                                 |
| 2026-07-13 | docs(thesis): voice and flow pass; fix EB limit in CTRW appendix                                          |
| 2026-07-13 | docs(thesis): revise dataset chapter after review notes                                                   |
| 2026-07-09 | test: pin p-dependence with invariants instead of a derived constant                                      |
| 2026-07-09 | test: strengthen conservativeness check; cover the ~ expansion branch                                     |
| 2026-07-09 | test: cover altitude-noise statistics/PSD aggregation and load_config                                     |
| 2026-07-09 | test: decouple suite from editable .pth; fix stale Config() calls                                         |
| 2026-07-09 | refactor(config): remove paraglider-first bias from load_config                                           |
| 2026-07-07 | logbook: seventh audit entry (global sweep)                                                               |
| 2026-07-07 | audit(global): consistency sweep across every document                                                    |
| 2026-07-07 | logbook: sixth audit entry                                                                                |
| 2026-07-07 | preproc(final): every hyperparameter pinned; tau_c measured; flight filter corrected on census evidence   |
| 2026-07-07 | logbook: fifth audit entry; unbreakable identifier in C2 pointed at the repro table                       |
| 2026-07-07 | preproc(audit): implementer's walkthrough — four logical bugs closed; C2 de-walled                        |
| 2026-07-07 | logbook: fourth audit entry                                                                               |
| 2026-07-07 | preproc(audit): measure the gap cap on the census; suspect-wait flag; blueprint realigned                 |
| 2026-07-07 | logbook: third audit entry                                                                                |
| 2026-07-07 | preproc(audit): 2*dt floor on the gap cap; interpolated mask; interior-ground detector; per-segment gates |
| 2026-07-07 | logbook: second split audit entry                                                                         |
| 2026-07-07 | preproc(gaps): absolute cap on bridgeable gaps; split mechanics made concrete                             |
| 2026-07-07 | logbook: four July entries; Next steps renumbered to the merged layout                                    |
| 2026-07-07 | preproc(split): state what a split means for each observable; fix the segment gate                        |
| 2026-07-07 | style(voice): sweep appendices B and C2                                                                   |
| 2026-07-07 | style(voice): sweep the body chapters                                                                     |
| 2026-07-07 | style(voice): sober headings and prose in the dataset chapter                                             |
| 2026-07-07 | impl(params): the fix-level working defaults as a table                                                   |
| 2026-07-07 | thesis(enu): signpost the two-step conversion                                                             |
| 2026-07-07 | impl(validation): the five cleaning checks as a numbered list                                             |
| 2026-07-07 | thesis(structure): fold data acquisition into the dataset chapter; let the text breathe                   |
| 2026-07-07 | cleaning(frozen-lock): rank the witness; quantify the body; lighten tab:cleaning                          |
| 2026-07-07 | ffvl link fixed                                                                                           |
| 2026-07-07 | thesis(cleaning-spec): one contract across body, appendix and blueprint                                   |
| 2026-07-07 | reporting(census): generate the census-scan numbers as \StatScan* macros                                  |
| 2026-07-07 | config(preproc): condition the vertical savgol timescale on alt_source; fix stale refs                    |

| Date       | Event (commit subject)                                                                                                         |
|------------|--------------------------------------------------------------------------------------------------------------------------------|
| 2026-07-07 | cleaning: make fix-level cleaning conservative — never lose a search/climb fix                                                 |
| 2026-07-07 | fix(altitude-PSD): correct the baro-vs-GNSS noise claim; robust median+band figure                                             |
| 2026-07-07 | bibliography: ground the transport-analysis & CTRW methods                                                                     |
| 2026-07-07 | bibliography: ground the pre-processing methods (Savitzky–Golay, Welch, WGS84, Hampel/MAD)                                     |
| 2026-07-07 | bibliography: verify every entry, archive sources locally, fix the FAI citation                                                |
| 2026-07-07 | polish(prose+typography): tighten cleaning sections, clear all overfull boxes                                                  |
| 2026-07-07 | logbook: record the fix-level cleaning redesign                                                                                |
| 2026-07-07 | redesign fix-level cleaning: three-detector taxonomy, clear removal semantics                                                  |
| 2026-07-07 | fix(typography): float placement, long-identifier wrapping, A/V prose                                                          |
| 2026-07-07 | restructure thesis: narrative body + implementation-details appendix                                                           |
| 2026-07-06 | fix(gap-fig): tighten Fig 4.6 panel (b) x-range; pin all figures in place                                                      |
| 2026-07-06 | fix(gap-fig): linear axis/bins for Fig 4.6 panel (a), reveals real small-integer structure                                     |
| 2026-07-06 | fix(sampling-fig): discrete bins for Fig 4.7 (native dt), fraction not density                                                 |
| 2026-07-06 | fix(fixlevel-fig): shared 13 m/s vertical-speed bound; fix a general long-caption LaTeX limit                                  |
| 2026-07-06 | fix(fixlevel-fig): integer-aligned bins for vz; revise horizontal-speed bounds to 45/55                                        |
| 2026-07-06 | fix(fixlevel-fig): widen x-range for vxy/vz tails; explain vz’s non-empty valleys                                              |
| 2026-07-06 | feat(preproc): per-discipline fix-level speed bounds, on a much larger sample                                                  |
| 2026-07-06 | fix(altitude-noise): Fig 4.1 panel (a) back to dual y-axis, over the full 30-min window                                        |
| 2026-07-06 | fix(thesis): Fig 4.1 redesign (drift over jitter), Appendix B panel-letter bug, missing sample sizes                           |
| 2026-07-06 | chore: add MIT LICENSE, README badges, and a real test-running CI                                                              |
| 2026-07-06 | ci(docs): auto-deploy on push again, scoped to paths that feed the site                                                        |
| 2026-07-06 | docs(logbook): reformat narrative notes for readability; re-sync Next steps; docs deploy on demand                             |
| 2026-07-06 | fix(altitude-noise): dual-axis + decoupled windows for Fig. 4.1 panels (a)/(b); chore(deps): dev/analysis as uv default groups |
| 2026-07-06 | docs: data-driven fix-level section; four-panel altnoise caption; blueprint, logbook, README                                   |
| 2026-07-06 | chore(data): explicit SSD->repo seasons_index sync; document the offline snapshot                                              |
| 2026-07-06 | feat(altitude-noise): four-panel figure (raw channels, difference, PSD, availability)                                          |
| 2026-07-06 | feat(preproc): data-driven fix-level diagnostics; gap-histogram readability; drop redundant time-gap                           |
| 2026-07-06 | fix(igc): treat only full-day drops as a midnight roll-over                                                                    |
| 2026-07-06 | chore: SSD data layout (raw/catalog/derived) + pyarrow for the scan cache                                                      |
| 2026-07-05 | thesis ch.4: data-driven pre-processing pipeline + geodesy figures                                                             |
| 2026-07-02 | Symmetric para/hang dataset; adopt barometric altitude; IGC parser + noise diagnostics                                         |
| 2026-07-01 | Rename soaring-ffvl -> soaring-para; add hang-glider data and update all docs                                                  |

| Date       | Event (commit subject)                                                                         |
|------------|------------------------------------------------------------------------------------------------|
| 2026-07-01 | Fix delta config: season start 2001, data_root /Volumes/SSD_DISANTE/hang_glidern/delta_cfd_igc |
| 2026-07-01 | Add soaring-delta CLI for hang-glider CFD data (delta.ffvl.fr)                                 |
| 2026-07-01 | Add analysis sub-package and preprocessing diagnostics; revise thesis chapters 1-4             |
| 2026-07-01 | Expand pre-processing: explicit cleaning and filtering sections                                |
| 2026-07-01 | Clarify stationarity/compatibility check paragraph                                             |
| 2026-07-01 | Add preliminary characterization section before main observables                               |
| 2026-07-01 | Add isotropy test protocol to the propagator item                                              |
| 2026-07-01 | Rewrite propagator item: formal derivation of marginal scaling                                 |
| 2026-07-01 | Reorganize TA-MSD item for clarity and typographic structure                                   |
| 2026-07-01 | Clarify ergodicity breaking and add CTRW appendix                                              |
| 2026-06-28 | Revise chapter 4 observables and the IGC description                                           |
| 2026-06-28 | Warn when thesis and logbook next-steps drift apart                                            |
| 2026-06-28 | Align logbook next steps with the thesis planning chapter                                      |
| 2026-06-28 | Publish the logbook alongside the thesis                                                       |
| 2026-06-28 | Add living thesis document and build automation                                                |
| 2026-06-26 | Deploy docs via GitHub Pages Actions artifact instead of gh-pages branch                       |
| 2026-06-26 | Fail fast with a clear message when data_root is unset                                         |
| 2026-06-26 | Add 'verify' subcommand and remove machine-specific data_root path                             |
| 2026-06-26 | Link published docs in README; drop Development section                                        |
| 2026-06-25 | Add GitHub Pages deployment via gh-pages branch                                                |
| 2026-06-25 | Translate all documentation, comments, and strings to English                                  |
| 2026-06-25 | Initialize thesis repo: FFVL CFD .igc data acquisition                                         |

## 2 Narrative notes

*Section and figure numbers inside dated entries refer to the document layout at the time of writing; the thesis has since been restructured (the dataset chapter is now Ch. 2, “Next steps” Ch. 3).*

**2026-06 — Paraglider data acquisition complete.** The full FFVL paraglider CFD archive (`parapente.ffvl.fr`, seasons 1999–2000 to 2025–2026) has been scraped: about 203,000 flight records, of which ~186,000 carry a GPS track. All 186,052 available tracks were downloaded.

**What worked** Impersonating a Chrome TLS fingerprint (`curl_cffi`) reliably passes the FFVL Cloudflare challenge; the resumable, atomic, validated download survived many interruptions without corruption.

**Gotchas** 173 tracks could not be obtained — 172 because the server returns something that is not a valid IGC (broken or placeholder links on FFVL’s side, so a retry does not help), and 1 due to a transient Cloudflare challenge (worth a later retry). On macOS/exFAT the OS scatters `._*` sidecar files next to every written file; a `clean` step removes them.

**2026-06 — Integrity verified.** A full on-disk re-scan (`verify`) confirmed that all 186,052 downloaded files are structurally valid IGC: no truncated files, no leftover `.part`, no read errors. The verification logic was promoted from a throwaway script to a first-class `soaring-para verify` subcommand (reusing the download-time validator), with tests.

**2026-06 — Robustness of the data path.** The external disk remounted under a different name (`HDD_DISANTE` → `SSD_DISANTE`), which would have silently broken a hard-coded path. Fix: removed every machine-specific path from the repo (placeholder + `SOARING_PARA_DATA_ROOT` environment variable) and added a start-up guard that aborts with a clear, actionable message when `data_root` is unset/unmounted — and verified that `/Volumes` is not user-writable, so a wrong path can never cause a silent download to the wrong place.

**2026-07 — Hang-glider data acquired; naming corrected.**

**Second source added** The hang-glider CFD (`delta.ffvl.fr`, seasons 2001–2002 to 2025–2026), sharing the acquisition code (`soaring.acquisition.ffvl`) via a second CLI entry point `soaring-delta` (config `delta_download.yaml`, env var `SOARING_DELTA_DATA_ROOT`). Data: ~9,300 flights, 6,716 tracks downloaded, 30 failures (broken server-side links, same pattern as paragliders). Stored at `/Volumes/SSD_DISANTE/hang_gliders/delta_cfd_igc/`.

**Naming corrected** The paraglider CLI was renamed from the generic `soaring-ffvl` to `soaring-para` (and `SOARING_FFVL_DATA_ROOT` → `SOARING_PARA_DATA_ROOT`), and its config from `ffvl_download.yaml` to `para_download.yaml`, to avoid ambiguity now that both sources are present.

**2026-07 — Hang-glider catalog built; datasets reorganized symmetrically.** Ran the full pipeline on the hang-glider data (`soaring-delta clean, verify, build-catalog, status`), which so far had only been downloaded: 9,259 flights, 6,746 with a track, 6,716 downloaded and all verified (0 integrity failures), 30 unrecoverable server-side links.

**Data layout** The local `data/` folder now mirrors the SSD — one directory per discipline (`paragliders/`, `hang_gliders/`), each with a versioned `seasons_index.csv` and a git-ignored `catalog.csv`.

**Thesis symmetry** `generate_stats.py` now reads both indices and emits per-discipline macros (`\StatPara*`, `\StatHang*`) and two per-season tables; the first three thesis chapters were made symmetric across both disciplines.

**2026-07 — Altitude channel decision: barometric.** Decided to take the vertical dynamics from the *barometric* altitude rather than the GNSS altitude, diverging from the reference pipeline (Jérémie’s code derives  $z$  and  $v_z$  from `AltGNSS`, reading but not using `AltPre`).

**Rationale** The pressure altitude has sub-metre relative precision and low high-frequency noise, its errors being slow (offset, drift) and killed by differencing, whereas the GNSS vertical carries a high-frequency noise floor that derivatives amplify. Confirmed empirically with a new noise diagnostic — an IGC B-record parser (`soaring.analysis.igc`) and an altitude-noise analysis (`soaring.analysis.altitude_noise`) whose Welch power spectra show the GNSS floor two to three orders of magnitude above the barometric one near Nyquist (thesis appendix on the method).

**Consequences** The A/V validity flag is subsumed by the missing-altitude check on the adopted channel (a V fix keeps position + baro on a baro flight, but is dropped on a GNSS-fallback flight); one altitude channel per trajectory, no baro/GNSS splicing; flights with no pressure sensor fall back to unfiltered GNSS, tagged by a per-flight `alt_source` column; the trimming threshold is on horizontal speed only ( $v_{xy} > 5$  m/s, sustained); cleaning/trimming run on the raw geographic coordinates (great-circle speeds), with the ENU conversion done *last*, origin at take-off.

**2026-07 — Baro-absence fraction: sized sampling, not a full sweep.** A full-population parse of the paraglider archive (186,052 files) for the baro-channel-absence figure took an

unacceptable  $\sim 45$  min serially; parallelising across 8 processes cut this to  $\sim 15$  min, still too slow to run routinely.

**Fix** A properly justified *simple random sample*: the standard proportion sample-size formula ( $n = z^2 p(1-p)/e^2$ , conservative  $p = 0.5$ ) gives  $n = 2401$  for a target 95%-confidence margin of error of  $\pm 2$  percentage points (`required_sample_size` in `soaring.analysis.altitude_noise`, with `proportion_ci` reporting the point estimate and interval). The population is treated as effectively infinite: at  $n = 2401$  out of 186,052 flights, the finite-population correction would move  $n$  by only  $\sim 30$ , not worth the extra parameter. The hang-glider archive (6716 files) stays a full census (cheap enough, under two minutes). Total runtime:  $\sim 50$  s. Result: paragliders  $(28.4 \pm 1.8) \%$  (95% CI, sampled), hang gliders 17.5 % (exact, census).

**Also caught mid-flight** The IGC parser accepted syntactically well-formed but semantically invalid B records (e.g. minute  $\geq 60$ , hour  $\geq 24$ , out-of-range latitude/longitude) without rejecting them; added explicit format-validity checks (9 new tests) and corrected the thesis wording, which had conflated this *implemented* format-validity layer with the fix-level *dynamics-plausibility* layer of Table 4.1 – at the time designed (a dataclass of thresholds) but not yet implemented as an enforced filter.

**2026-07 — Pre-processing made data-driven and config-driven.** Reworked Chapter 4 so the flight-level filtering is computed from the *full parsed dataset* (paragliders and hang gliders), not the declared catalog metadata.

**What changed** A track scan (`soaring.analysis.census.scan_tracks`) parses every downloaded track; Figure 4.2 shows the real duration and flown-path-length distributions with per-variable *marginal* retention curves. Adopted cuts: duration  $\geq 40$  min and path length  $\geq 30$  km (a minimal cut,  $\sim 95\%$  retained); the earlier min-fix-count cut was dropped as redundant with duration, and the *extent* proxy was replaced by total flown *path length* (median 86 km para, 158 km hang).

**Key decision** Every pre-processing threshold now lives in `configs/preprocessing.yaml` (never hard-coded), loaded via `load_preproc_config` into typed dataclasses.

**Also added** Great-circle (haversine) inter-fix speed/distance *before* the ENU conversion; a Savitzky–Golay smoothing/differentiation section with a noise-matched procedure for its two hyperparameters; the switch from a common resampling cadence to native per-flight  $\Delta t$  (Figure 4.5); and an implementation blueprint (`docs/guide/preprocessing-pipeline.md`) recording the exact schema and steps, to retire into code once the pipeline is built.

**2026-07 — Gap diagnostics; scan cache; baro-presence consolidated.** Added Figure 4.5, the sampling-regularity diagnostics (largest gap relative to native  $\Delta t$ , missing fraction of a uniform grid, each with its marginal retention curve), fulfilling a promise Sec. 4.1.6 already made but had not yet delivered.

**Scan cache** A fresh full parse would have repeated the  $\sim 15$ – $20$  min paraglider census just done for Figure 4.2, so the per-flight scan is now **cached** to `data/<discipline>/track_scan.parquet` (gitignored, like `catalog.csv`; `load_or_scan_tracks` reads it if present, else scans and writes it — later moved to the SSD, see below). `track_stats` was extended with a few near-free byproducts of the same scan — barometric-presence fraction, max horizontal/vertical speed, barometric altitude range — for future validation of Table 4.1’s fix-level bounds.

**Consolidation** The barometric-presence fraction (Sec. 4.1.1) now reads from this same cache (`altitude_noise.baro_presence_from_scan`) instead of running its own separate scan, turning the paraglider number from a sized sample into an exact census at no extra cost: 30.1 % (exact,  $n = 186,025$ ) — consistent with the earlier sampled estimate,  $(28.4 \pm 1.8) \%$ .

**Also caught** A pre-existing fragility in `altitude_noise.collect` where a single unreadable

file (a transient external-disk hiccup) aborted the whole PSD sample; it now skips and logs the file instead.

## 2026-07 — SSD reorganised; no data kept locally; figure fixes.

**Policy going forward** Raw data, catalogs and every analysis byproduct live only on the external SSD, never in the local repo.

**Reorganisation** Each discipline’s `data_root` is now laid out by maturity — `raw/{igc,raw_xml}` (untouched acquisition output), `catalog/{catalog.csv,seasons_index.csv}` (tables derived from it), `derived/` (further byproducts, today just `track_scan.parquet`; future cleaned/filtered data and analysis results get their own `processed//analysis/` siblings), with `Config` updated accordingly (new `derived_dir` property). Verified file counts before/after the move (paraglider: 186,052 tracks, 27 XMLs; hang glider: 6,716 tracks, 25 XMLs — all matched) and that both acquisition CLIs still work against the new layout. Previously-local `track_scan.parquet` copies moved to `<data_root>/derived/`; orphaned local `catalog.csv` copies (superseded by the track-based scan months ago) were removed.

**Figure 4.5 (gaps)** The adopted `max_gap_factor` was raised from 5 to 10 — at 5 it retained only 77.8%/65.0% of para/hang flights (nowhere near the near-full retention of the duration/path cuts), at 10 it reaches 86.9%/85.1%. Considered 20 (90.8%/90.6%, little further gain) and rejected it: the cut is *relative* to each flight’s own native  $\Delta t$ , so for the slower hang-glider loggers (up to 10 s native) a factor of 20 would tolerate interpolating across a  $\sim 200$  s gap — long enough to span a thermal climb. Also tightened the distribution panels’ axis range to the data’s own percentiles (the old 0–200/0–0.6 range was mostly empty).

**Figure 4.1 (altitude noise)** Panel (a)’s difference trace now uses a colour distinct from both channels (was reusing GNSS’s red); its window grew from 10 to 30 minutes; a Nyquist-frequency line was added to panel (b) (previously discussed in the caption but never drawn). Also found and fixed a real bug in the “representative flight” selection: picking the *longest flight by elapsed time* broke on a 30-hour, 1,100-fix outlier (one corrupted timestamp gap) — fixed to select by *fix count* instead, which has no such failure mode.

**Local data folder clarified** The two versioned `seasons_index.csv` are the offline reproducibility anchor for the thesis’s dataset counts (the document builds with no SSD mounted), not stray local data. Their sync from the canonical SSD copy is now explicit and automatic (`refresh_seasons_index.py`, run best-effort by the pre-commit hook), so the snapshot cannot silently drift.

## 2026-07 — Fix-level cleaning made data-driven; parser roll-over bug; figure and data-folder cleanups.

**Fix-level cleaning audited** Table 4.1 is now audited on the data, like the flight-level cuts: a new figure in Sec. 4.1.2 shows the per-*fix* distributions of horizontal speed, barometric vertical speed and barometric altitude (`make_fixlevel_diagnostics_figure`, from `fix_level_distributions` over a seeded sample — millions of fixes). What justifies a fix-level bound is the fraction of *fixes* it removes, not the number of flights it touches; the per-flight *maxima* already in the scan (`max_vxy_mps` and friends) answer the wrong question, so they do not place the cuts.

**Gap handling unified** Removed the absolute 60 s inter-fix time-gap from the fix-level table: it duplicated the sampling-regularity cut of Sec. 4.1.6, which is more principled (relative to each flight’s own native  $\Delta t$ ). Neither was ever an enforced filter, so this is a spec simplification, not a behaviour change.

**Parser bug fixed** Found the root cause of the “30-hour, 1,100-fix” pathology worked around previously: the IGC parser’s midnight roll-over unwrap treated *any* backward step in time-

of-day as a new day and added 86,400 s, so a few-second GPS glitch became a spurious multi-hour gap. Now only a drop of (nearly) a whole day counts as a roll-over, with residual backward jitter clamped to keep the series monotonic. Added regression tests for both the true roll-over and the glitch.

**Figure and config cleanups** The altitude-noise figure (Sec. 4.1.1) is now four panels — (a) raw channels, (b) their difference, (c) PSD, (d) fallback rate — fixing the old a/b/d labelling. The gap-diagnostics figure (Sec. 4.1.6) no longer clips its tail into an artificial edge spike, with an x-range fitted to the data’s own percentiles. Result-specific numbers (retention percentages, one-off anecdotes) were pruned from YAML/code comments — such numbers belong in the thesis, not in reusable config.

**Local data folder clarified** Rewrote `data/README.md` and fixed several stale references to a flat `data/seasons_index.csv` path that never existed.

**2026-07 — Coordinate/geodesy figures.** Redrew the coordinate-transformation figures: Figure 4.3 (WGS84 ellipsoid; geodetic vs geocentric latitude) and Figure 4.4 (the ECEF→ENU rotation, in our notation), and added the ESA Navipedia citation for the ECEF conversion.

**2026-07 — Altitude-noise figure (Fig. 4.1): panels (a)/(b) redesigned.** A closer look at the rendered figure (not caught by any test, since nothing there errors) found two compounding problems in the one-flight time-series panels, both traced to a single shared 30-minute window: the several-hundred-metre climb dominated the vertical scale, hiding the GNSS jitter the panel exists to show; and panel (b), subtracting only the scalar *mean* of the GNSS–barometric difference, ended up on a  $\pm 20$  m scale that looked inconsistent with the  $\sim 130$  m gap plainly visible in panel (a).

**First attempt (reverted)** Shortened the shared window to 2 minutes and switched panel (b) to a *linear* detrend (matching `_welch`’s). This resolved both symptoms but destroyed something worth keeping: the GNSS–barometric difference actually drifts by tens of metres over a full flight (plausibly the barometric channel’s departure from the ICAO standard atmosphere as altitude changes) — a real, physically meaningful feature that a 2-minute window is too short to show and a linear detrend actively removes.

**Final design** Decoupled the two panels’ windows and mechanisms instead of forcing one compromise on both. Panel (a) keeps a short (2-minute) window but now plots barometric and GNSS on *two independent y-axes* (same numeric span, different baseline) rather than one shared axis, which previously had to stretch from one channel’s minimum to the other’s maximum — wasting on blank space exactly the room needed to compare the two channels’ roughness; with matched spans the GNSS trace is visibly more jagged than the barometric one. Panel (b) reverts to the original 30-minute window with only the *mean* removed (no detrend), so the slow drift is visible again alongside the residual jitter, with the removed mean offset (139 m for this flight) annotated directly on the panel. Both panels’ wording was generalised from “climb” to “vertical motion”, since a short window can just as well land on a glide.

**2026-07 — dev/analysis moved from pip extras to uv default dependency groups.** Chasing the figure fix above, a bare `uv sync --reinstall-package soaring` (no `--extra` flags) silently reset the environment to the core-only dependency set, uninstalling `pytest/ruff/mypy/matplotlib/scipy/pyarrow` — and, on the way, hit a pre-existing uv bug (uninstall failing on packages whose RECORD file was missing) that required rebuilding `.venv` from scratch.

**Root cause** `dev` and `analysis` were plain PEP 621 optional-dependencies, so *any* `uv sync/uv run` invocation without matching `--extra` flags resyns the venv down to exactly what was requested, silently dropping whichever extras the previous invocation had added —



alternating flagged and unflagged commands across a session means constant install/uninstall churn.

**Fix** Moved both to `[dependency-groups]` (PEP 735) with `[tool.uv] default-groups = ["dev", "analysis"]`, so every `uv sync/uv run` includes them unconditionally. Neither group was ever meant to be pip-installed by an external consumer in practice (this repo is not published), so the loss of `pip install .[dev]`-style installability is immaterial; `docs` remains a real extra since it is genuinely on-demand. The docs-deploy CI workflow now opts out of the new default groups (`-no-default-groups`) to stay minimal.

**Docs updated** `README.md`, `docs/guide/installation.md` (dependency-groups table, pip fall-back instructions), the two reporting scripts' now-unnecessary `uv run -with matplotlib -with scipy` incantations, and the `soaring.analysis` package docstring.

**2026-07 — Logbook readability; Next-steps drift; docs deploy on demand.** Three unrelated loose ends, tidied together.

**Logbook typography** The narrative notes below were dense, single-paragraph entries — hard to scan. Reformatted the longer ones into short lead sentences plus labelled sub-points (*what changed*, *why*, *gotchas*), and dropped a stale forward-reference in the altitude-channel entry (“to confirm on a larger sample once the SSD is remounted”, superseded two entries later by an exact 186,025-flight census).

**Next-steps drift** The Next steps list below had fallen out of step with the thesis chapter it is meant to mirror: *Preliminary characterization* (thesis Sec. 4.2) and *Tooling* (Sec. 4.9) were missing entirely, and the whole-trajectory, phase-resolved and transport-analysis bullets were missing content the thesis already has (the propagator/return-probability/first-passage observables; the step-waiting coupling and phase-sequence memory; the stratification and bias-control plan). Re-synced against the current chapter.

**Docs deploy** The GitHub Pages workflow redeployed on every push to `main`, including pushes with nothing to do with the site (e.g. acquisition/analysis code with no docstring change) — not wanted. One such deploy had also failed with a generic, transient “try again later” from GitHub’s Pages API, but that was an isolated glitch, not a config problem (confirmed via the Actions log: only that one run failed out of the last ten, and only at the deploy step, after a successful build). Fix: the `push` trigger is now scoped with `paths` to what actually feeds the built site (`docs/**`, `mkdocs.yml`, `src/soaring/**`, whose docstrings `mkdocstrings` renders into the API reference), so the site still updates itself automatically but only for changes that matter to it; `workflow_dispatch` stays available for an on-demand redeploy.

**2026-07 — License added; README badges; a real test-running CI.** `pyproject.toml` had declared `license = MIT` since the project’s start, but no `LICENSE` file existed, and the repo had no workflow that actually ran the test suite (only the docs-deploy one).

**License** Added `LICENSE` (MIT): permissive, minimal text, the de facto standard for research Python code, and consistent with what `pyproject.toml` already declared – no reason to pick something more restrictive for code meant to be freely reused.

**New CI: tests.yml** Runs `uv run pytest` with coverage on every push to `main` and every PR. Needed no `-extra` flags to get `pytest-cov/matplotlib/scipy/pyarrow` – a direct payoff of making `dev/analysis` `uv` default groups earlier. Coverage XML uploads to Codecov as a best-effort step (`fail_ci_if_error: false`): a Codecov hiccup, or the repo not being linked there yet, must never redden the Tests badge.

**README badges** Added five badges (python version, Tests workflow status, Codecov coverage, docs-online link, license) at the top of `README.md`. The Codecov one needs a one-time manual step outside this repo – linking `matteodisante/soaring-anomalous-transport` on

codecov.io (GitHub login) and, if it asks for one, adding CODECOV\_TOKEN as a repo secret – which cannot be done from here.

**2026-07 — Fig. 4.1/4.2 and Appendix B: closer reading, several fixes.** A line-by-line pass over the altitude-noise figure, the fix-level figure, and the PSD appendix, prompted by direct questions about what they actually show.

**Fig. 4.1 panel (a), third look** The dual-axis, short-window version (see the entry above) did not read, on the page, as clearly showing GNSS is the noisier channel, so it was dropped for a shared axis over the full 30 min window instead – which showed the offset drift directly, but on request the panel went back to a dual axis once more, this time over that same 30 min window rather than a short one. Net result: each channel is independently auto-scaled (barometric left, GNSS right), so the two curves track almost on top of each other (both are the same  $\sim 500$  m climb, just at a different baseline) – good for reading each channel’s own shape, but it means the raw vertical gap is no longer a literal reading of the offset. That number was never really this panel’s to carry: it lives in panel (b), and the noise-floor claim in panel (c), regardless of which of these three versions panel (a) went through.

**Appendix B (PSD) bug** Two cross-references pointed at “Figure 4.1(b)” for the PSD, which has been panel (c) since the four-panel redesign; fixed. Also added the actual flight yield of the  $\Delta t$ -matching step, which the text described only qualitatively: of 400 sampled flights per discipline, 212 paragliders but only 114 hang gliders pass it, because hang gliders split across several native rates while paragliders are overwhelmingly 1-second loggers – so the hang-glider PSD curve is honestly an average over that 1-second minority, not the full hang-glider population, now stated as such.

**Fig. 4.2 (fix-level), added** The exact sample (800 flights per discipline, 6,633,923/4,541,511 paraglider/hang-glider fix pairs) was missing from the caption; added. Also explained two things a reader flagged as confusing on sight: panel (a) (horizontal speed) shows a knee well below the adopted 40 m/s bound ( $\sim 23/35$  m/s for para/hang) – read as the transition from ordinary gliding speeds to rarer but real high-energy manoeuvres (wind-assisted glides, steep spirals) rather than GPS error, hence the bound is deliberately kept above it, not moved down to it, so as not to discard genuine dynamics; and panel (b) (vertical speed)’s comb-like histogram is a quantisation artefact (integer-metre barometric altitude differenced over a 1-second native step can only return an integer m/s), not a plotting bug.

**Table 4.1, added** A one-sentence justification for the  $-100$  m lower altitude bound, which read as arbitrary: it is not a claim about true terrain elevation, but headroom for the barometric channel’s own non-QNH-corrected offset (Figure 4.1 measures this directly,  $\sim 100$  m for the flight shown there), so a genuine sea-level take-off can log slightly negative without being cut.

**2026-07 — Fix-level speed bounds made per-discipline, on a much larger sample.** Chasing the loosened-threshold experiment of the previous entry to a conclusion.

**Sample size** FIXLEVEL\_SAMPLE\_PER\_DISCIPLINE raised from 800 to 15,000: 15,000 paragliders (126,723,499 fix pairs,  $\sim 73$  s) and all 6,716 hang gliders (a full census at this population size, 37,915,334 pairs,  $\sim 25$  s) – under two minutes total, and it resolves the sparse tail well enough to distinguish real dynamics from a rare artefact, which the 800-flight sample could not.

**The hang-glider bump, confirmed and explained** At this sample size the 45–90 m/s feature is unambiguous: not a smoothly decaying tail but a distinct, much more slowly-decaying plateau (confirmed with a fine, 2.5 m/s-binned histogram – the decay rate visibly changes between 37.5 and 42.5 m/s, from a steep real decline to a near-flat plateau lasting to  $\sim 90$  m/s). A sustained *ground* speed of 160–324 km/h has no credible airborne explanation for a hang glider: read as a logger/GPS artefact population, not rare-but-genuine manoeuvres.

**Decision: per-discipline bounds, not one shared value** Paragliders and hang gliders have

different performance envelopes, confirmed by their different tail shapes – a shared bound is either too loose for one or clips real dynamics of the other. `FixLevelThresholds.max_horizontal_speed_mps`, `max_vertical_speed_mps` are now `dict[str, float]` keyed by discipline (`configs/preprocessing.yaml`, a nested mapping; a "sailplanes" entry can be added later with no code change). Vertical speed: 12 m/s (para) / 14 m/s (hang).

**Horizontal speed bound, revised again: 40 was still “slowing”, not “stopped”** First adopted at 25/40 m/s (para/hang), where the fine-binned decay rate first visibly changes. On review, with the panel’s  $x$ -range widened (next entry) to actually show the far tail, that point turned out to still be on the shoulder of the decline, not the flat artefact itself: the bin-to-bin ratio only settles at  $\approx 1$  (a genuine plateau, decay actually stopped) around 55 m/s for hang gliders, and the analogous point for paragliders is  $\approx 45$ , not 25. The 25/40 m/s pair was therefore cutting into the tail-end of real (if rare) dynamics – a wind-assisted glide, a steep spiral – rather than sitting cleanly past it. Revised to **45 m/s (para) / 55 m/s (hang)**, both now Table 4.1’s and the config’s adopted values.

**Figure and table updated** `make_fixlevel_diagnostics_figure` now draws one dashed cut per discipline, colour-matched to its histogram, with a per-discipline removed-fraction annotation (previously one shared cut and fraction). Table 4.1 gained a row per discipline for the two speed bounds.

**–100 m altitude bound, empirical check (unchanged, kept for the record)** Computed the per-flight mean GNSS–barometric offset over 1500 flights per discipline (broader than the single Figure 4.1 example): median  $\approx 90$  m, 25th–75th percentile roughly 43–135 m. This is a *whole-flight* average, and the offset itself grows with altitude (Sec. 4.1.1), so it likely *overstates* the offset specifically near the ground; confirms the order of magnitude of –100 m without pinning it down precisely. No error found in Figure 4.2 itself.

**$x$ -range widened for (a)/(b)** The panels’ range was cropping the view almost exactly at the cut (the 99.9th percentile of the pooled sample sits right there, by construction), hiding how much – and what – the bound leaves out. Panels (a)/(b) now extend to each discipline’s own 99.99th percentile (computed per discipline, not pooled, so hang gliders’ heavier tail is not swamped by paragliders’ larger sample); panel (c) (altitude) keeps the original, narrower range on purpose – its cuts sit far out in an otherwise tight, unimodal distribution, and widening it the same way mostly shrinks the informative part of the panel.

**Panel (b)’s non-empty valleys, explained and checked** The comb (spikes at integer m/s) was expected; that the valleys *between* integers were not exactly zero was not, and looked like a bug. It is not: not every flight samples at 1 s (Figure 4.7 – hang gliders spread across several native rates), and differencing over a longer native step returns a finer-grained set of values (e.g. multiples of 0.2 m/s at a 5 s cadence) that fill the gaps once every cadence is pooled. Checked directly, split by native rate: among 1 s-native fixes,  $\sim 99\%$  land within 0.05 m/s of an integer (the comb, essentially pure); among every other native cadence, only 34–42% do – the other 58–66% is exactly what shows up between the teeth.

**Panel (b)’s jagged comb, fixed** Separately from explaining *why* the valleys were not empty: the comb itself looked bad – an arbitrary, fine (linspace) grid of bins straddles the integer spikes inconsistently depending on where each bin edge happens to fall, which reads as noisy up-down jitter rather than the smooth decaying envelope the data actually has. Fix: one bin per integer, centred on it (edges at half-integers) rather than an arbitrary edge grid – `make_fixlevel_diagnostics_figure` now takes an `integer_aligned` flag, set for panel (b) only ( $v_{xy}$  is not similarly quantised). The panel is now a clean, monotonically-decaying curve; the removed-fraction numbers are unaffected (computed from the raw data, not the histogram).

**Vertical speed reverted to one shared bound: 13 m/s** Unlike horizontal speed, paragliders’ and hang gliders’  $|v_z|$  distributions sit close enough together (panel (b)) that a per-

discipline split buys nothing; a single 13 m/s bound (between the previous 12/14 split) covers both, with one shared grey dashed line and one pooled removed-fraction annotation, mirroring how the altitude band already works. `FixLevelThresholds.max_vertical_speed_mps` reverted from `dict[str, float]` to a plain `float`; the figure’s two shared-cut helpers (band and single-cut) were merged into one `_shared_cut_panel` that handles either, so panel (b) still gets the `integer_aligned` treatment above.

**A LaTeX limit hit, and fixed for good: very long captions** Rebuilding after the above changes hit ! `Dimension too large` / “I can’t work with sizes bigger than about 19 feet”, deep in `\@makecaption`. Bisected the Figure 4.2 caption by half repeatedly to isolate it – not a typo (braces and math-mode  $\$$  were balanced throughout), but the caption’s sheer length (~3900 characters): plain L<sup>A</sup>T<sub>E</sub>X’s single-line-fit check for captions measures the *whole, unwrapped* caption as one hbox before deciding whether to wrap it, and past a few thousand characters that measurement alone overflows T<sub>E</sub>X’s ~19-foot maximum dimension, even though the wrapped result would render fine. Loading the `caption` package does the same check by default (to centre genuinely short captions), so `singlelinecheck=false` was needed on top of it. This is a general limit of this document now that several figures carry long, data-justified captions – not specific to this one – so the fix (`main.tex` preamble) covers all of them going forward, not just Figure 4.2.

**2026-07 — Figure 4.7 (native sampling interval): discrete data, discrete plot.** Raised independently: the density in Figure 4.7 rose above 1, and native  $\Delta t$  is a discrete quantity, so why plot a density at all. Checked on the full census:  $\Delta t$  lands on an exact whole second for 99.9% of paragliders and 100% of hang gliders – essentially a categorical variable (one of a handful of configured logger rates), not a continuous one. The previous plot used arbitrary 0.5 s-wide bins on this discrete data (`np.arange(0.25, 8.75, 0.5)`): most paragliders’ mass concentrated in the single bin straddling  $\Delta t = 1$ , and `density=True` divides a large fraction by that narrow bin width, which is exactly how it exceeded 1 – a real number, not a bug, but a genuinely confusing one for data this concentrated. Fixed the same way as the vertical-speed comb: bins one full second wide, centred on each integer (1, ..., 10, plus a  $\geq 11$  bin for the long thin tail), so each bar reads directly as “fraction of flights at exactly this rate.” With unit-width bins `density=True` is numerically identical to that fraction (dividing by a width of 1 changes nothing), so the axis is now labelled *fraction of flights*, not *density* – accurate either way, but only one of the two labels says what is actually being counted. Real distribution, now stated precisely in the caption instead of the vaguer “1, 2, 5,  $\geq 8$ ” before: paragliders overwhelmingly (69%) at 1 s; hang gliders spread across 1 s (30%), 5 s (27%), 10 s (15%), 2 s (13%), and a long thin tail beyond 11 s – a fuller picture than the old caption’s “up to 8 s,” corrected in Sec. 4.1.6 too.

**2026-07 — Figure 4.6 (gap diagnostics), panel (a): dropped the log axis.** Same family of question as the two entries above, this time about the largest-gap-to- native- $\Delta t$  ratio: is it also discrete, why a density, and (specifically here) is a log  $x$ -axis actually needed.

**Not as clean-cut as native  $\Delta t$**  Checked directly: only 86% of paraglider and 63% of hang-glider values are exact integers (thousands of distinct values in all, tail out to  $> 50,000$ ) – a missed fix or two at the native rate is the single most common cause of a large gap (hence the pull towards small integers), but plenty of gaps are not a clean multiple of it. So, unlike Figure 4.7, this is not purely categorical; integer-only bins would have thrown away a genuine continuous component.

**The log axis, though, was not earning its keep** Panel (a) only ever displays the *bulk* of the distribution (up to `gap_hi`, a modest  $\sim 1$ –20 here) – the heavy tail is deliberately left to the retention panels (c)/(d), which already sweep the full range on a log axis, justified there since the cut spans orders of magnitude. Over the narrow, already-cropped range panel (a) actually shows, a log axis buys nothing and actively hides structure: log-spaced bins are

extremely narrow near 1, so the large mass sitting at small integers – a real, common failure mode, not noise – gets compressed into a sliver.

**Fix** Panel (a) switched to a linear axis with linear, evenly-spaced bins over the same `gap_hi`-bounded range. This did not just fix the axis choice: it revealed a real pattern that the log/narrow-bin combination had been washing out – visible peaks at small-integer ratios, most clearly for hang gliders (around  $k = 3, 6, 10, 15$ ), consistent with the “missed native-rate fixes” mechanism. Panel (c)’s log axis is unaffected and still the right choice there, for the opposite reason: it has to show a cut ranging over orders of magnitude.

**2026-07 — Why gap ratios aren’t integers: stratified by native rate.** Follow-up on the panel (a) entry above: if a missed native-rate fix is the single most common cause of a large gap, why is the largest-gap ratio an exact integer only 86%/63% of the time, not effectively always? Stratifying the full census by native  $\Delta t$  answers it.

**$\Delta t = 1$  s is trivially always integer** Every IGC timestamp already has whole-second resolution, so the largest gap itself is always a whole number of seconds; dividing it by  $\Delta t = 1$  cannot produce anything but an integer. Checked directly: 100% for both disciplines at this rate. Since 1 s is also the dominant paraglider rate (69% of flights, Figure 4.7), it alone pulls the paraglider-wide fraction up to 86%.

**Slower configured rates are only about half integer** At  $\Delta t = 2/5/10$  s the integer-ratio fraction is 60%/58%/51% for paragliders and 34%/61%/39% for hang gliders – no rate stands out, none close to 100%. The mechanism from the panel (a) entry still holds (a missed fix is still the main driver of a large gap), but the gap itself is a real elapsed duration – GPS reacquisition time, logger buffering, a signal dropout – set by whatever caused the interruption, not by the logger’s configured cadence. Nothing requires that duration to land on a multiple of  $\Delta t$ ; it does so about half the time by chance, not by construction.

**So the headline 86%/63% is a mixture effect** Paragliders are predominantly 1 s loggers (trivially integer), hang gliders are spread across several slower rates (mostly not) – the two overall percentages mainly reflect which rate dominates each fleet, not some fleet-wide regularity in how gaps occur.

**2026-07 — Figure 4.6, panel (b): tightened the missing-fraction axis.** The  $x$ -axis ran out to roughly twice the adopted cut (0.10), leaving the right half of the panel empty: the 90th percentile of the pooled missing fraction sits at  $\approx 0.09995$ , essentially at the cut itself, so nothing of substance lives beyond it. Changed the margin past the cut from  $2\times$  to  $1.1\times$ , i.e. `miss_hi = max(p90, cut * 1.1)`, giving a 0–0.11 range that still clears the 90th percentile with a little headroom but no longer wastes half the panel on empty axis.

**2026-07 — Figures now stay where they are placed in the source.** Plain  $\text{\LaTeX}$ ’s float algorithm was free to defer any figure past its ideal spot to the next page it judged to have room, and with several large figures placed close together in Sec. 4.1 it was pushing some many pages away from the text discussing them – annoying for anyone reading top to bottom rather than jumping via cross-references. Added the `float` package and switched every `figure/table` placement specifier (previously `[t]/[!]`) to `[H]` (“here, literally”), which pins each one to its position in the source, full stop. No layout regressions on a full rebuild: text and figure land on the same page throughout, including the two cases checked directly (Figures 4.6 and 4.7, immediately after the paragraphs that introduce them).

**2026-07 — Thesis restructured: narrative body, implementation appendix.** The body had grown heavy: captions and prose mixed physical/statistical reasoning with code module names, exact config values, sample sizes and computation times (worst case, Fig. 4.2’s

caption, 3898 characters). Restructured so the body reads like an ordinary thesis and every implementation-level fact moves to one place, cross-referenced tersely from where it used to sit.

**New back-matter part.** “Implementation and Computational Details,” a document-within-the-document at the very end: unnumbered part divider (a numbered “Part I” would have implied a nonexistent “Part II”), its own local “Contents of this part” listing (`titletoc`), and its own independent chapter numbering (plain arabic, reset – continuing the A/B appendix lettering would have made it read as “just more appendices”). Mirrors the body chapter by chapter: Overview, Data acquisition, The dataset (incl. pre-processing and preliminary characterization), Next steps.

**One macro, used throughout.** `\impldetails{impl:x}` renders “(implementation details: Sec. X)” wherever a fact was trimmed out of the body; `impl:*` labels mirror `sec:*` 1:1.

**Chapter 3/4 boundary moved.** Trajectory pre-processing and preliminary characterization are dataset description, not planned analysis – moved from “Next steps” into “The dataset” (which now runs to Sec. 3.6); “Next steps” starts at “Observables on the whole trajectory.” Five cross-references that would otherwise have become circular self-references were updated in the same pass.

**Every pre-processing figure/table slimmed.** Fig. 4.2’s caption: 3898 to ~1500 characters, keeping the artefact argument (hang-glider plateau, physically-absurd km/h translation) and moving exact sample sizes/timings/bin mechanics. Table 4.1, Figs. 4.1/4.6/4.7 similarly. `B-psd.tex`: three clauses extracted from “Welch’s method” (segment length, `scipy` name, overly specific detrending framing), its whole “Computational details” section relocated, one genuine external-validity caveat (hang-glider PSD averages a 1 s-logger minority) preserved in “Reading the altitude spectra.” `02-data-acquisition.tex`: CLI/subcommand names, atomic-write/retry mechanics, env-var names relocated; two hardcoded approximate flight counts fixed to use the `\Stat*` macros instead (drift risk).

**One Python edit.** The two season-table captions (`scripts/reporting/generate_stats.py`) had their provenance clause replaced with the macro; regenerated, `ruff` clean.

Verified throughout: every `impl:*` label matches an `impldetails` call (no dangling cross-reference), full rebuilds clean at each stage (no undefined references), official build script (`scripts/build_docs.sh thesis`) green.

## 2026-07 — Fix-level cleaning redesigned: three detectors, unambiguous removal.

A critical re-reading of the fix-level cleaning, asking whether the adopted design (Table 3.6.2 / old 4.1) is actually reliable, exposed real gaps and turned the flat list of absolute bounds into a design organised by *how* each defect presents. Cleaning only, not the flight-level filtering.

**The gap that mattered** “Reject the fix” was ambiguous on the one question that decides everything: when the speed between two fixes is too high, *which* of the two is removed? A speed lives on the segment between fixes and cannot say. And a fixed global speed bound is only a teleport filter — it passes the moderate spikes (a 30 m/s step for a 12 m/s paraglider) that dominate the differentiated velocity/acceleration, and is blind to a frozen GPS lock (repeated position, zero speed, clock still running), which mimics a long CTRW waiting time and would plant false trapping events in the very tail the anomalous exponent is read from.

**Guiding principle** Cleaning must not bias the transport statistic it precedes: assume only that the trajectory is smooth at the sampling scale, never a model of the motion. This rules out the tempting “best” upgrade, a Kalman / robust motion-model smoother — its constant-velocity or diffusive prior would write the studied behaviour into the data.

**Three detectors, by context** *Off the trend* (spikes): a robust local-outlier test (Hampel — median centre, MAD scale, threshold in robust  $\sigma$ , plus an absolute floor  $\varepsilon_{\min}$ ), a per-*node* quantity, which names the offending fix directly and dissolves the which-fix ambiguity. *On the*

*trend* (structural): duplicate time  $\rightarrow$  centroid; non-roll-over backward time  $\rightarrow$  delete; frozen-lock run  $\rightarrow$  mark as gap. *Out of range*: the old absolute bounds, kept as the interpretable backstop the figure already audits.

**Two removal semantics** Split apart what “remove” meant: a corrupt position/time deletes the node (gap bridged later at resampling); a corrupt altitude invalidates only the altitude channel, keeping the horizontal position — so the primary (horizontal) MSD loses nothing to a barometric spike. The old table coupled the two, throwing away good  $(x, y, t)$  for an altitude fault.

**Frozen lock  $\rightarrow$  split, not bridge** A lost-lock run is marked as a data gap and the flight is *split* there into independently analysed segments — not interpolated across (which invents motion) nor dropped whole (which wastes the good sides). Folded into the uniform-sampling step, which already floated segmentation as a refinement.

**Validation, both directions** The existing “removes a negligible fraction of fixes” argument only bounds false *positives*. Added two checks for the other direction: an injected-defect / recovery test (planted spikes/freezes/time-glitches of known size give the detection rate, i.e. the false negatives, and calibrate the parameters) and a downstream-invariance test (MSD exponent and waiting-time tail versus cleaning strength; adopt the plateau where they stop moving — the evidence that error, not signal, was removed).

**Where it lives** Narrative in Sec. 3.6.2 (Table regrouped by family with an explicit action column, fix-level figure reframed as the backstop audit); full working definitions — Hampel parameters, frozen-lock thresholds, single-pass  $O(N)$  anchor scan, integrity gate, validation protocol — in the dataset chapter of the implementation appendix. The cleaning routine and its new config keys  $(w, k, \varepsilon_{\min}, \varepsilon, \tau_{\text{freeze}}, f)$  remain to be built; the blueprint (`preprocessing-pipeline.md`) and its open items were updated to match. Thesis compiles clean (50 pp, no undefined refs, no new overfull boxes).

**2026-07 — Frozen-lock witness ranked; the cleaning spec quantified.** A consistency audit of the frozen-lock rule found a real flaw: the V/zero-alt flags sat in a plain or-branch and could overrule a climbing barometer, while the IGC standard says V certifies the GNSS *altitude*, not the horizontal fix.

**Fix** The witness of `eq:frozenlock` is now ranked and case-split by `alt_source`: on a barometric flight, baro-flat or byte-identical repeats (the only baro-independent override, since no moving receiver rewrites identical bits); V/zero-alt decide only on GNSS-fallback flights. All four contract locations updated together (thesis body, implementation appendix, blueprint, `preprocessing.yaml` comment).

**Guarantee quantified** A real climb misses the cut on two independent instruments:  $2R \approx 40\text{--}100\text{ m}$  against  $\varepsilon \approx 10\text{--}15\text{ m}$ , and at least 60 m of barometric gain over  $\tau_{\text{freeze}}$  against a few metres of  $\delta_z$ . The one assumption (a working barometric channel) is now stated, with the residual corner cases confined to the GNSS-fallback minority.

**Presentation** Hampel identifier named and cited in the body; the speed bound reframed from “backstop” to corroboration gate, matching the appendix; `tab:cleaning` regrouped with an invariant row; the seven detector parameters collected in a working-defaults table (`tab:fixlevel-params`); thresholds stated inline where each decision is made (the prose said “a few  $\Delta t$ ” where the adopted bound is  $10 \Delta t$ ).

**2026-07 — Acquisition folded into the dataset chapter.** Data acquisition is no longer a chapter: it opens “The dataset” as Sec. 2.1 (the file is `\input` there), its implementation mirror folded into the dataset mirror the same way, and the introduction roadmap rewritten; the thesis now has three chapters. In the same pass the pre-processing order (i)–(vii) became an enumerate mapping the chapter, the trimming rule became an equation ( $t_{\text{on}}/t_{\text{off}}$  as min/max of sustained

$v_{xy} > v_0$  windows), and the implementation appendix’s worst walls of text were rewritten in recipe form (Hampel test as numbered steps, detector order and the five validation checks as lists).

**2026-07 — Voice pass: the document must read human.** The prose had accumulated a generated-sounding register, partly from the AI-assisted sessions themselves: narrativized headings (“One rule: clean without biasing the transport”), aphorism setups, meta-commentary (“the ranking is the point”), a courtroom metaphor sustained across the cleaning section, and dense em-dash chains. Swept the whole document: headings are now nominal (Design principle, Removal policy, Frozen-position runs), the cleaning narrative rewritten plainly with content and numbers unchanged, the metaphor kept only where it is notation (the witness  $W$ ), a duplicated variometer sentence removed, prose dash pairs roughly halved. Appendix A already read plain and was left untouched.

**2026-07 — Split semantics pinned down.** Asked directly whether splitting a flight at a long gap can bias the analysis. The split itself is the right call (interpolating fabricates motion, dropping the flight wastes data), and ensemble observables are safe by construction: a segment keeps its parent’s origin and clock, and GPS positions are absolute, so  $\mathbf{r}(t)$  after the gap is valid and the flight merely contributes nothing at the missing lags. Three cautions were previously unstated and are now in the spec: phase durations truncated at a segment boundary are censored, not complete (else every split reads as a spuriously short wait and thins the  $\psi(\tau)$  tail); a segment is an aged observation window, not a fresh flight, for the time-averaged MSD; and the flight-level cuts are *not* re-applied to segments — the earlier draft said they were, which would have preferentially discarded the data behind late-flight gaps — replaced by a minimal segment-duration gate (config key with the routine).

**2026-07 — Second split audit: the gap bound gains an absolute scale.** Two questions from a design review, before implementation. (1) The gap bound was purely relative ( $10\Delta t$ ): a 3 s logger could bridge 27 s, a 10 s logger 100 s, while the harm of a bridged gap is set by the motion’s timescales, not the logger’s (an interpolated stretch is a straight line; near the thermalling period it can replace a full circle and read downstream as a glide). The split bound is now  $\min(10\Delta t, 20\text{ s})$ : the cap is at least twice the slowest common cadence (one missed fix never splits), at most about two thirds of a thermalling period, and well below  $\tau_{\text{freeze}}$  (an excised frozen run is split, so a native gap of that length cannot be bridged). Added as `max_gap_seconds` to the YAML and the `SamplingThresholds` dataclass, with the config test extended. (2) How a split works mechanically: the ENU conversion precedes it, so both segments already carry parent-referred coordinates and time; a split assigns a `segment_id` in the `fixes` table, per-segment records go to a `segments` table — no separate files, and the ensemble MSD reads  $\mathbf{r}(t)$  wherever a segment covers  $t$ . Also flagged: trimming strips only the *outer* ground phases, so a mid-flight landing and relaunch in one log would read as an implausibly long wait; an interior-ground guard (sustained near-zero  $v_{xy}$  of order ten minutes, treated like a long gap) is planned with the implementation.

**2026-07 — Third audit: interpolation budget, interior landings, per-segment gates.** (1) Interpolation error over a bridged gap is governed by the unresolved manoeuvring, not by the interpolant order: during a glide the error over 10–20 s stays at metres (below GPS noise), while during circling a gap of  $g$  seconds spans an arc of  $2\pi g/T$  that no interpolant can recover — at  $g = 10\text{ s}$  the chord-arc deviation is already  $\approx 0.5R$  (10–25 m), at  $g = 20\text{ s}$  up to  $2R$ . This confirms the two-scale bound and adds a refinement: the cap is floored at  $2\Delta t$  ( $g_{\text{max}} = \min(10\Delta t, \max(20\text{ s}, 2\Delta t))$ ), so the thin tail of very slow loggers also tolerates a single missed fix (a bridged gap there inserts at most two grid points). Every filled grid point is now



flagged `interpolated` in `fixes`, so phase statistics can treat reconstructed stretches with the same caution as segment boundaries. (2) The interior-ground detector is specified, built like the frozen-lock rule (cut only on corroborated evidence, since a wing soaring into wind holds near-zero ground speed while flying):  $v_{xy} < v_0$  sustained for  $T_{\text{ground}}$  (order ten minutes) *and* barometric altitude flat within a few metres; a detected stint is excised and handed to the split machinery. (3) Two gate corrections: the integrity gate counts over the airborne window only (ground junk that trimming removes anyway must not condemn a flight), and the missing-fraction gate acts per segment (a bad segment is dropped alone). Savitzky–Golay edge handling pinned (`mode='interp'`, per segment).

**2026-07 — Fourth audit: the gap cap measured on the census; blueprint realigned.**

With the SSD mounted, the two-scale gap bound was measured on the cached census scans (paired  $\Delta t$ /largest-gap per flight, thresholds read from the config, never hard-coded): the combined bound leaves 15.3% of paraglider and 40.5% of hang-glider flights with at least one split, against 11.8%/14.5% under the relative rule alone. The increase sits at the slow cadences ( $\Delta t \geq 2$  s), where a 20 s hole is only a few missed fixes. A split is cheap for ensemble observables but fragments within-segment statistics, so the cap stays a working value and enters the downstream-invariance sweep; the shares are generated census macros (`StatScan*GapSplit*`) quoted by the thesis. Blueprint realigned with the thesis: the stale “no second absolute bound is needed” sentence removed, `segment_id/interpolated` added to the `fixes` schema, a `segments` table added, Savitzky–Golay per-segment `mode='interp'` recorded, open items refreshed ( $\delta_z$ , `min_segment_duration`, `T_ground`, suspect-wait flag), and the header now states the working rule: implement from the blueprint, thesis normative on conflict. Sub- $T_{\text{ground}}$  stints are now *flagged* rather than merely auditable: the suspect-wait flag (near-zero  $v_{xy}$  with flat barometric trace, any duration) feeds  $\psi(\tau)$  sensitivity re-fits. README refreshed: `soaring.analysis` exists, and the logbook is committed with the repo, not git-ignored as the old text claimed.

**2026-07 — Fifth audit: implementer’s walkthrough; C2 de-walled.** Traced one IGC file through every step of the spec as an implementer would, asking at each point “what exactly does the code do here?”. Four real logical bugs surfaced, all fixed across the contract:

**Backward-time attribution** “Delete the fix with the corrupt time” did not say which one; a running-max rule would let a single forward-jumped clock condemn every later fix. Now: minimal removal (complement of the longest increasing subsequence).

**Position discontinuity** A re-acquisition offset (track jumps and stays) fits neither the spike model (no bounded block restores the trend — deletion would cascade) nor the frozen-lock model. Now its own row in `tab:cleaning`: split at the step, like a long gap; both sides are genuine, only the transition is unknown.

**`interpolated` was undefined under resampling** Every grid point is technically interpolated once a flight is resampled. Now: measured iff a fix lies within half a native step of the grid time; the same tolerance defines `was_resampled`.

**Savitzky–Golay window collapse**  $w = \text{odd}(\tau_c/\Delta t)$  falls below a cubic fit’s minimum for slow loggers ( $\Delta t$  of order  $\tau_c$  or above). Now floored at  $w = 5$ , with the over-smoothing caveat stated and those flights already stratified out of  $\Delta t$ -sensitive observables.

Smaller closures:  $\delta_z$  got a number ( $\approx 5$  m); Hampel needs a minimally populated window (order five fixes, else absolute bounds only); the frozen-lock  $\Delta z$  witness is median-based so an uncleaned altitude spike cannot fake a climb; suspect-ground stints are stored as per-flight intervals (phases do not exist yet at pre-processing time); a flight with no sustained-speed stretch is discarded with reason; duplicate uploads added to the blueprint’s open items. Readiness verdict: the pipeline is implementable end-to-end today up to smoothing; the three Savitzky–Golay  $\tau_c$  remain the one true blocker (placeholders pending the ENU PSD study; procedure and code path

exist). Remaining C2 walls of text (uniform-sampling thresholds, segment handling, flight-filter diagnostic sample, PSD band reading, flag-to-action) rewritten as labelled lists, same content.

## 2026-07 — Sixth audit: every hyperparameter pinned; the flight filter re-measured.

Goal: implementable through step (vii) included, no ranges left.

$\tau_c$  **measured, blocker closed** New script (`estimate_savgol_timescales.py`): Welch on  $E, N$  (local equirectangular) and on the altitude channel each flight uses, seeded 1 Hz sample (908/719/189 flight-channels). All three knees sit at  $f_c \approx 0.2 \text{ Hz} \Rightarrow \tau_c = 5 \text{ s}$ : every channel's typical floor is IGC quantization (measured  $0.60/0.26 \text{ m}^2/\text{Hz}$  against theoretical  $0.57/0.17$ ), not receiver noise. The baro/GNSS asymmetry lives in the spectrum's level, not its knee; even p90 spectra move it by less than a second. Placeholders replaced with measured values.

**Flight filter was cutting the wrong thing** Census forensics: above 40 min, path < 10 km catches 6 flights of 184,583 while the adopted 30 km cut removed 5,229 genuine localized flights (median extent 7.4 km) — the most-trapped tail, whose exclusion would bias the transport toward travellers. The declared minimal-threshold strategy itself dictated lowering: `min_path_km` 30  $\rightarrow$  10. Also found and fixed: fifteen “flights” of 16–166 h (loggers left running, paths to 10,000 km) now cut by an upper duration bound (16 h); a few hundred retained records with total altitude range under 100 m (vehicle logs or stuck sensors) now caught by an altitude-activity floor (75 m) plus a baro-liveness check at channel choice (range < 30 m  $\Rightarrow$  fall back to GNSS — a stuck baro previously passed as a baro flight with  $v_z \equiv 0$ ).

**All working ranges pinned** Hampel ( $w = 20 \text{ s}$ ,  $k = 5$ ,  $\varepsilon_{\min} = 15 \text{ m}$ , min. window 5 fixes), frozen-lock ( $\varepsilon = 15 \text{ m}$ ,  $\delta_z = 5 \text{ m}$ ,  $\tau_{\text{freeze}} = 60 \text{ s}$ ),  $f = 0.10$ , segment gate 90 s, interior-ground (600 s, 5 m): all now YAML keys with dataclass fields and config-test assertions. The injected-defect calibration owns the final values.

**Figures regenerated** The pre-processing diagnostics were rebuilt from the SSD with the new thresholds (the path-cut line moved to 10 km).

**2026-08 — The pipeline built, and the first MSD audited.** Stages (i)–(vii) implemented end to end and run over both archives: 155,788 paraglider flights retained of 186,052 (83.7%) and 6132 of 6716 hang-glider ones, written as `fixes/segments/flights_meta` and checked by `verify_dataset.py` (every invariant holds) and `check_reproducible.py` (500 sampled flights re-run through the code and identical). One bug found by the verifier's own blind spot: reading Parquet row groups instead of whole flights split every flight straddling a boundary, over-counting flights by 0.7% and truncating exactly the longest segments; `stream_flights` is the single correct reader that resulted.

Then the two MSD estimators, and an audit of what they mean. The audit is a second streaming pass keeping each flight's position at every lag, because the questions worth asking of an averaged curve are reductions of a per-flight quantity that no longer exists once the average is taken. What it found changed how the result is reported: the ensemble curve is *not* a power law (residual 15.2% and 31.2% rms against 3.8% and 3.6% for the time-averaged one), its local slope runs 1.31–2.21 and goes super-ballistic, and that shape survives fixed cadence, fixed duration and the removal of the common heading. A slope above 2 needs a ballistic component switching on at a finite time, and the first crossing of 5 km from take-off has median 1911 s — the scale at which the overshoot peaks. So the exponent is read off the time average and the ensemble curve yields a timescale; the aging reading, which compared two fitted slopes, is withdrawn, since one of them is not an exponent.

Sec. 2.8 built with it: retained-ensemble statistics, take-off maps on real Natural Earth coastlines (three frames — metropolitan France, La Réunion, and the world for the rest), and two results that constrain Ch. 3. The ensemble is anisotropic ( $\langle E^2 \rangle / \langle N^2 \rangle = 0.80$ , flat over three decades), so the 1-D marginal cannot stand in for the 2-D propagator; and the strata split by

wing class (40%) and terrain (62%) but not by season (12%), so the archive may be pooled over time and not over class or terrain. Verified: 383 tests, macro contract green, `regenerate.sh` end to end reproducing every quoted number, thesis compiling at 99 pages with no errors.

**2026-07 — Seventh audit: global consistency sweep.** Closing pass over every document at once, hunting statements left behind by the day’s changes rather than new design flaws. Thirteen stale spots found and fixed: ranges still quoted where values are now pinned ( $k$ ,  $w$ ,  $\varepsilon$ ,  $\tau_{\text{freeze}}$ , in body and appendix); “joins the config when the routine is built” where the keys now exist; the blueprint carrying two open items that contradicted each other about the same parameters; “two track quantities” where the filter now has three criteria; the repro-table caption counting two figure scripts where there are four; the logbook’s own Next-steps still calling the timescales placeholders. One reconciliation of substance: the body’s smoothing section predicted asymmetric windows (smooth barometer, noisy GNSS) and the measurement contradicted it — the knees coincide, the asymmetry lives in the spectrum’s level — so the text now states the expectation, the measurement, and why the per-channel conditioning stays. The measured-timescales script joined the reproducibility table. Verified end to end: no stale grep hits, mkdocs builds, 78 tests pass, thesis and logbook compile with no errors, no undefined references, no overfull boxes.

**2026-06 — Documentation.** Project docs are built with MkDocs and published on GitHub Pages. Initially deployed with `mkdocs gh-deploy`, which accumulated one built-site commit per deploy on a `gh-pages` branch; migrated to the official GitHub Pages *artifact* workflow (build → upload artifact → deploy), removing the branch and keeping the git history clean.

**Operational note.** In this repo `uv sync` occasionally fails to (re)install the editable package; `uv sync --reinstall-package soaring` fixes it.

### 3 Next steps

These mirror the planning chapter of the state document (`thesis/sections/04-next-steps`, itself titled “Next steps”) and are kept in sync with it, section by section. The near-term plan is to analyse the data *in parallel* with studying the segmentation, since many observables can be built before any phase splitting.

- **Data sources.** FFVL paraglider and hang-glider CFD data are collected, cataloged and verified. Extend to further sources – notably WeGlide for sailplanes (a higher-quota API request is pending; the default rate limit makes a bulk export infeasible) and possibly XContest – merging into one catalog and stratifying by source. Refresh the per-discipline `seasons_index.csv` via `build-catalog` so the document statistics stay current.
- **Pre-processing (thesis Sec. 2.7).** The parser is in place (`soaring.analysis.igc`) and the pipeline is designed, thresholded (`configs/preprocessing.yaml`) and audited on real data: fix-level cleaning (great-circle speeds),  $v_{xy}$ -based trimming, data-driven flight-level filtering (duration + path length), the ENU conversion (barometric vertical), native- $\Delta t$  uniformity, and Savitzky–Golay smoothing/differentiation. **Built and run** (2026-08): the seven stages are production code under `soaring.analysis.preproc`, driven by `scripts/preprocess.py` into the three Parquet tables, unit-tested, and checked against the invariants of Ch. 2 by `verify_dataset.py` and against the code itself by `check_reproducible.py`. The Savitzky–Golay timescales are measured (all knees at  $\approx 0.2$  Hz, so  $\tau_c = 5$  s). Remaining: the calibration battery for Hampel, frozen-lock and the integrity gate, via injected defects and downstream invariance.

- **Preliminary characterization (Sec. 2.8). Done (2026-08).** Trajectory statistics on the retained ensemble; take-off maps on real coastlines, three frames (metropolitan France 94.7%, La Réunion, the world for the rest). Two results that constrain the analysis rather than merely describe it. Isotropy *fails*:  $\langle E^2 \rangle / \langle N^2 \rangle$  sits at 0.80 for paragliders across three decades, so the 1-D marginal may not stand in for the 2-D propagator and the components are analysed separately. And the strata are *not* compatible by wing class (40% typical departure from the pooled MSD) or orographic group (62%), though they are by season (12%) — so the archive may be pooled over time, not over class or terrain. Remaining: displacement maps.
- **Whole-trajectory observables (Sec. 3.1, no segmentation).** Characterise the data with observables that need no phases. **Both MSD estimators are built and audited (2026-08):** a single exponent describes the time-averaged curve ( $\alpha_{TA} = 1.732$  and  $1.688$ , residual under 4%) and only summarises the ensemble one (15.2% and 31.2%), whose local slope crosses over — super-ballistic at the top — where the population leaves its launch area. So the ensemble curve yields a timescale rather than an exponent, and the ergodicity test cannot take the form of a gap between two fitted slopes; it moves to the EB parameter and the phase-resolved statistic, which compare distributions. Still to build: the one-dimensional marginal propagator and its scaling collapse; the return probability and the non-Gaussian parameter; the velocity autocorrelation and turning-angle distribution; first-passage/first-exit times; and the speed/vertical-velocity distributions. Goal: a first physical/statistical picture, independent of any phase splitting.
- **Segmentation (Sec. 3.2).** Study Jérémie’s internship report and code, then re-examine the transition/search/climb HMM (binary features: sign of vertical speed, curvature-angle persistence, straightness) — decide whether to keep that approach or add complexity for a better segmentation on our larger, more heterogeneous data.
- **Phase-resolved observables (Sec. 3.3).** After segmentation: the waiting-time decomposition (climb  $\tau_c$ , search  $\tau_s$ , and their sum  $\tau_w$ ), the glide step-length/duration joint law, per-flight anatomy (thermal count, phase time/distance budget), and the angular diffusion between successive glides. Plus two couplings that decide which transport model applies: the step-waiting coupling ( $\ell$  vs  $\tau_w$ , the Lévy-walk signature) and memory in the phase sequence (transition-matrix structure, duration correlations), which test the renewal assumption behind the textbook CTRW/Lévy-walk results.
- **Reproduce the reference (Sec. 3.4).** Check that the enlarged paraglider-and-hang-glider dataset reproduces Vilpellet et al.: per-phase conditional MSD (their Fig. 3) and per-phase descriptive statistics (their Fig. 4), plus the global  $H \approx 0.88$  — validation and universality test (the sailplane comparison becomes reachable once that source is in hand).
- **Minimal model (Sec. 3.5).** A minimal stochastic model for  $H \approx 0.88$ : resolve transition segments, merge search+climb into one effective state, with angular diffusion between successive transition directions (preliminary idea, to be refined/changed).
- **Tooling (Sec. 3.6).** Largely underway already: the analysis package mirrors acquisition, and the dataset statistics plus several diagnostic figures (pre-processing, altitude noise) already regenerate automatically from the data rather than being pasted in by hand. Remaining: extend the same treatment to whatever tables/figures the transport analysis produces, once it exists.